Predicting Customer Ratings of Beauty Products Based on Amazon Reviews

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Link to the video recording of the presentation:

https://drive.google.com/file/d/13UVpUAcbJJEIUE4-2sxq4pm5DOtbvrPA/view?usp=sharing

Research Questions

Motivation: Online customer reviews contain rich information about how users experience products, yet much of this information is unstructured and difficult to interpret at scale.

RQ 1: Can customer review text be used to **predict** whether a review corresponds to a high or low product rating?

- supervised machine learning task
- classification task
- Logistic Regression, Decision Tree, Random Forest

RQ2: What are the distinct rating **dimensions** that customers emphasize when reviewing beauty products?

- unsupervised machine learning task
- topic modeling task
- BERTopic

RQ3: How are individual products rated on each dimension based on the predicted rating and topics?

Data

Data source:

- Amazon Reviews 2023 dataset collected in 2023
- Use a subset of the review dataset for the All_Beauty category, with 14405 reviews posted in 2023
    - ensuring computational feasibility by reducing the sample size
    - optimizing the classifier for contemporary review analysis

Data

14405 samples, 7 features (4 text, 1 numerical, 1 categorical, 1 timestamp)

Text features:

- **title**: title of the review
- **text**: the review text
- **product_title**: the name of the product, 5397 unique values
- **parent_asin**: the primary identifier for the product, 5397 unique values

Numerical feature:

- **rating** (1-5): the corresponding rating of the product from the review
    - standard deviation: 1.56
    - mean: 3.87
    - median: 5

Data

14405 samples, 7 features (4 text, 1 numerical, 1 categorical, 1 timestamp)

Categorical feature:

- **verified_purchase**: 2 categories
    - True: the purchase of the review is verified
    - False: the purchase of the review is not verified

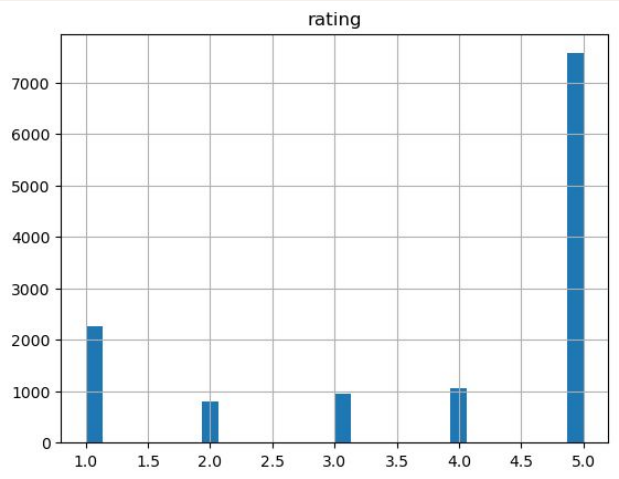
Timestamp feature:

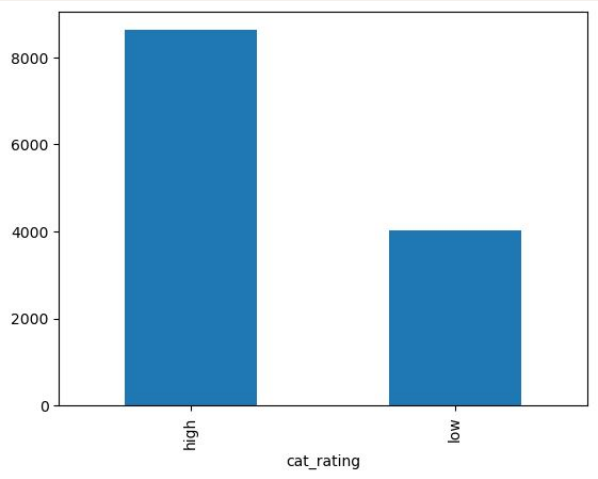
- **timestamp**: the date and time at which the review was posted, from Jan 1, 2023 to September 9, 2023.

Exploratory Data Analysis

Data Cleaning and Processing:

- remove missing values, duplicates, and unverified purchases
- **Target variable**: Transform “rating” into categorical feature **“cat_rating”**
  - 1-3 low (0): point out product flaws
  - 4-5 high (1): convey positive endorsement

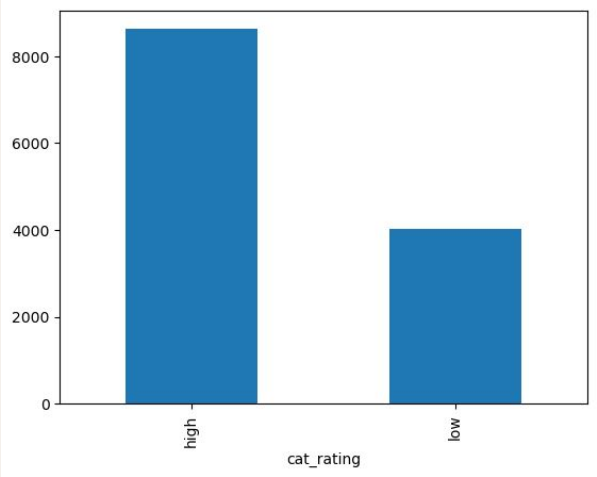


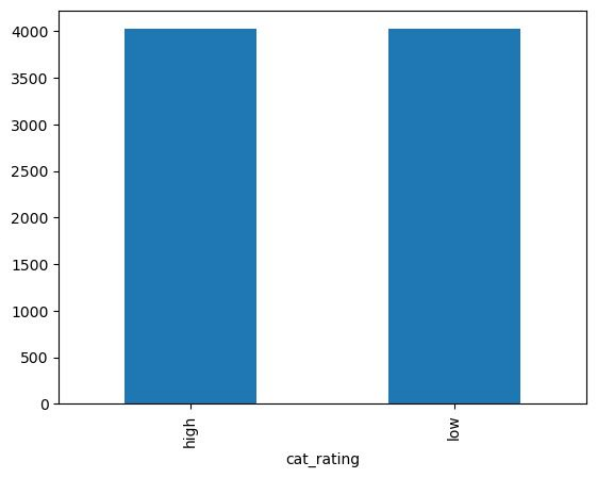


Exploratory Data Analysis

Data Cleaning and Processing:

- undersampling: balance the dataset to avoid prediction bias
- cleaned dataset: 8048 samples (4024 high, 4024 low)

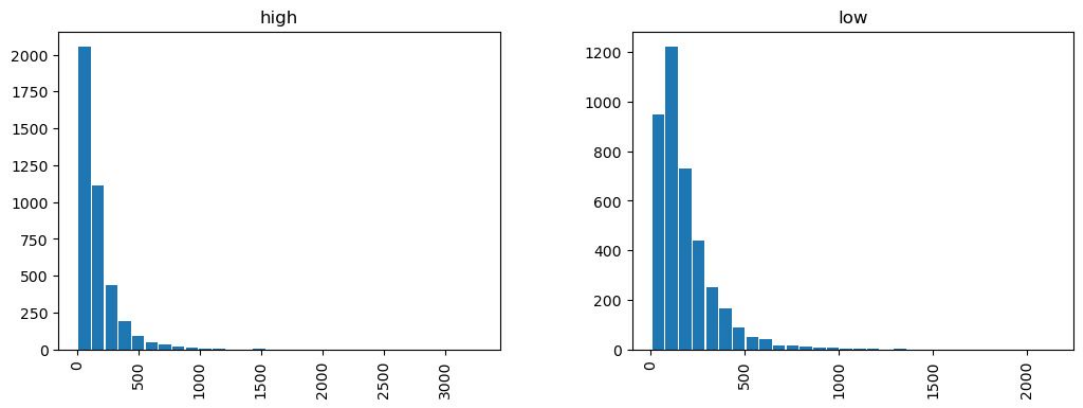




Exploratory Data Analysis

Data Cleaning and Processing:

- **Feature variable**: merge “title” and “text” as **“review”**
- Distribution of review length (4 - 3289 words)
  - mean: 172.8, standard deviation: 173.1
  - no clear difference between two classes



Exploratory Data Analysis

Data Cleaning and Processing:

- Text data cleaning and representation
  - expand contractions into normal words, lowercase, and remove html tags, non-alphabetic characters, url, punctuations, and repetitive letters
  - tf-idf vectorizer: 4101 features

Supervised Machine Learning

Model exploration and hyperparameter tuning:

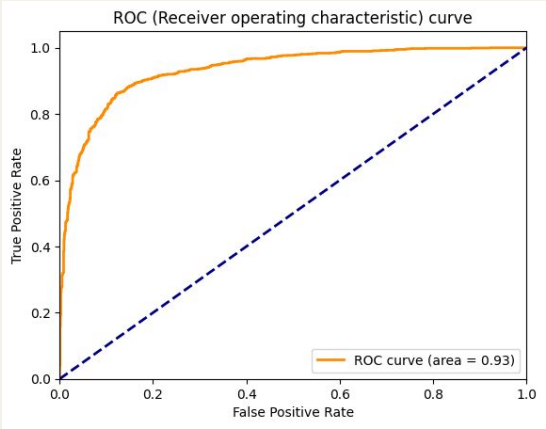
- Logistic Regression, Decision Tree, Random Forest
- split training and testing data: 70% training, 30% testing
- GridSearchCV: hyperparameter tuning for each model

Supervised Machine Learning

Logistic Regression Model:

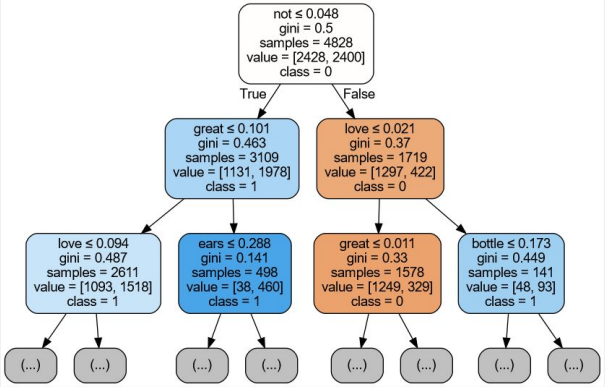
- L2 penalty, C = 1, lbfgs solver
- f1-score: 0.87
- AUC = 0.93
- recall (0): 0.89, recall (1): 0.85

	precision	recall	f1-score	support
0	0.85	0.89	0.87	1187
1	0.89	0.85	0.87	1228
accuracy			0.87	2415
macro avg	0.87	0.87	0.87	2415
weighted avg	0.87	0.87	0.87	2415



features	coefficients
great	6.23
love	6.01
amazing	4.28
perfect	3.85
best	3.49
nice	3.07
good	2.95
easy	2.83
well	2.64
excelente	2.56

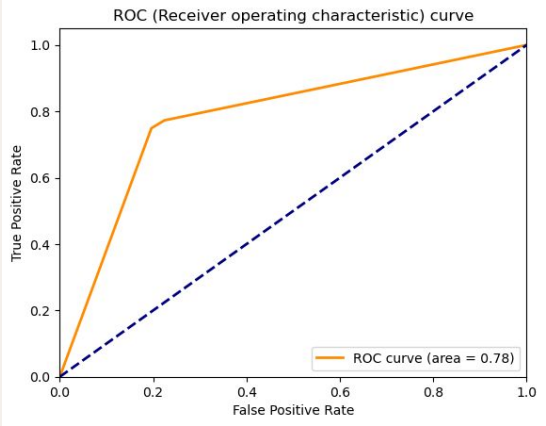
Supervised Machine Learning



Decision Tree Model:

- criterion: gini, max_depth: None, min_samples_split: 2, min_samples_leaf: 1
- f1-score: 0.78
- recall (0): 0.80, recall (1): 0.75
- AUC = 0.78

features	importances
not	0.16
great	0.06
love	0.06
good	0.02
perfect	0.02
best	0.02
amazing	0.02
favorite	0.01
easy	0.01
well	0.01



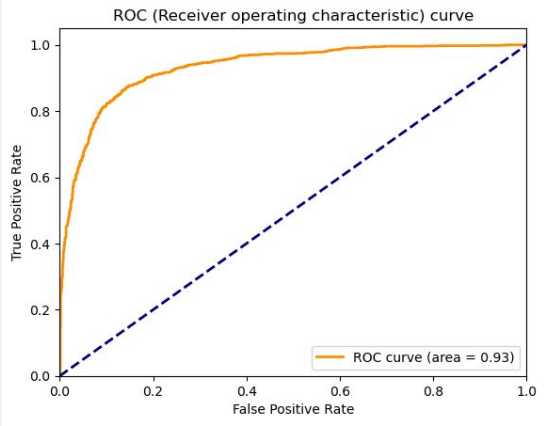
	precision	recall	f1-score	support
0	0.76	0.80	0.78	1187
1	0.80	0.75	0.77	1228
accuracy			0.78	2415
macro avg	0.78	0.78	0.78	2415
weighted avg	0.78	0.78	0.78	2415

Supervised Machine Learning

Random Forest Model:

- criterion: gini, n_estimators: 200, max_features: log2, max_depth: None, min_samples_leaf: 1
- f1-score: 0.86
- recall (0): 0.89, recall (1): 0.83
- AUC = 0.93

	precision	recall	f1-score	support
0	0.83	0.89	0.86	1187
1	0.89	0.83	0.86	1228
accuracy			0.86	2415
macro avg	0.86	0.86	0.86	2415
weighted avg	0.86	0.86	0.86	2415



features	importances
not	0.05
great	0.03
love	0.02
good	0.01
perfect	0.01
amazing	0.01
well	0.01
easy	0.01
best	0.01
no	0.01

Supervised Machine Learning

	f1-score	AUC	recall (0)	recall (1)	Top 5 features
Logistic Regression	**0.87**	**0.93**	**0.89**	**0.85**	**great**, **love**, amazing, **perfect**, **best**
Decision Tree	0.78	0.78	0.80	0.75	not, **great**, **love**, **good**, **perfect**
Random Forest	0.86	**0.93**	**0.89**	0.83	not, **great**, **love**, **good**, **perfect**

Supervised Machine Learning

Comparison:

- Logistic regression is the best performing model. It adopted L2 regularization, which reduces overfitting and improves generalization.
- The decision tree is prone to instability and overfitting, especially at greater depths.
- The random forest mitigates this weakness by aggregating many trees, improving robustness and reducing variance.
- Overall, the results suggest that ensemble methods and regularized linear models are both highly effective for sentiment classification, while a single tree is comparatively less reliable.

Supervised Machine Learning

Error analysis:

A common false positive error:

Silky **smooth**(1.143) but **slips**(-0.140) off The **material**(-0.328) is **great**(6.227) It is more **like**(0.073) a doo **rag**(0.173) or **wave**(0.101) **cap**(-0.429) It just **keeps**(1.468) **slipping**(-0.148) off while **im**(0.265) **sleep**(-0.124)

- The positive feature “great” has very high importance in the classifiers.
- The review expressed both positive and negative comments.
- The classifiers rely on individual words, unable to learn the context.

Unsupervised Machine Learning

Why BERTopic:

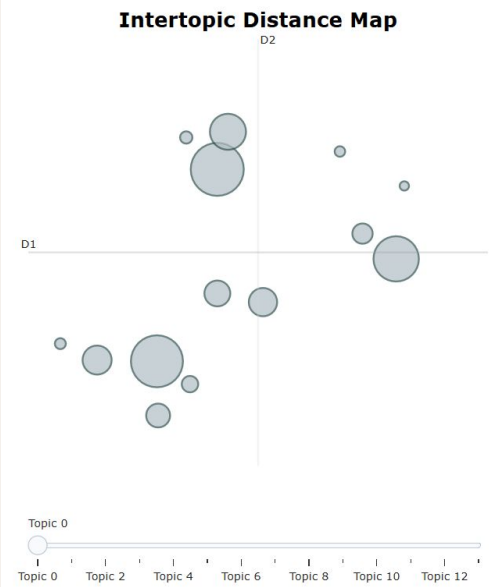
- The dataset consists of short, and informal product reviews.
- Traditional models such as LDA do not work well on short documents.
- BERTopic leverages contextual sentence embeddings and clustering to capture semantic similarity between reviews.

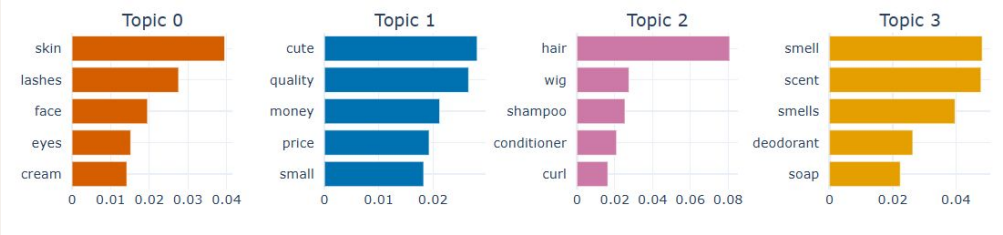
Unsupervised Machine Learning

Modeling:

- hyperparameter tuning:
  - CountVectorizer
  - UMAP: n_neighbors=15
  - HDBSCAN: min_cluster_size=30
  - nr_topics=15

- result: 14 topics
- evaluation: topic coherence: 0.47





Unsupervised Machine Learning

Topic ID	Topic Name	Top Keywords	Number of Reviews
0	product_skin_eye_face	skin, lashes, face, eyes, cream	1164
1	**dimension_size_quality_value**	**cute, quality, money, price, small**	**1120**
2	product_hair_care_wigs	hair, wig, shampoo, conditioner, curl	851
3	**dimension_fragrance_scent**	**smell, scent, smells, deodorant, soap**	**538**
4	**dimension_packaging_bottle_pump**	**bottle, bottles, pump, broken, opened**	**349**
5	product_nail_products	nail, nails, polish, gel, coat	332
6	product_hair_tools_clips_shavers	hair, clips, blades, comb, shaver	277
7	**dimension_color_shade**	**color, colors, purple, blonde, palette**	**235**

Unsupervised Machine Learning

Topic ID	Topic Name	Top Keywords	Number of Reviews
8	product_brushes_bristles	brush, brushes, bristles, dog, hair	170
9	product_glue_tape	stick, sticky, glue, tape, adhesive	111
10	product_dental_tools	teeth, dentist, plaque, dental, cleaning	63
11	**dimension_durability**	**broke, broken, apart, plastic, fell**	**50**
12	product_waxing_hair_removal	wax, waxing, hair, roll, legs	46
13	product_dryer_diffuser_fit	diffuser, dryer, hair dryer, orchid, fit	36
-1	unassigned	smell, hair, scent, color, quality	1707

How are individual products rated on each dimension

Product	Grucioso Gel Nail Polish Set	Sekaler Heated Eyelash Curler
size/quality/value	2.59	2.67
scent	/	/
packaging	4.39	/
color	4.27	/
durability	/	1.11
predicted rating	3.76	3.30

rating: 5pt

Conclusion

RQ1: Can customer review text be used to **predict** whether a review corresponds to a high or low product rating?

- Customer reviews can predict the high or low rating of a product with logistic regression classifier, decision tree classifier, and random forest classifier. The logistic regression classifier has the strongest prediction ability.

RQ2: What are the distinct rating **dimensions** that customers emphasize when reviewing beauty products?

- 5 out of 14 topics identified by BERTopic are about dimensions of products: 1) size, quality and value, 2) scent, 3) packaging, 4) color, 5) durability.

RQ3: How are individual products rated on each dimension based on the predicted rating and topics?

- By applying machine learning methods to review text, we can measure product performance across key aspects.